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To:
The Selection Committee
COLCOM Postdoctoral Position in Artificial Intelligence
College of Computing (COLCOM)
Mohammed VI Polytechnic University (UM6P)
Lot 660, Hay Moulay Rachid, 43150 Benguerir, Morocco

Subject: Application for the Postdoctoral Position in Artificial Intelligence (College of Computing)

Dear Members of the Selection Committee,

I am writing to express my enthusiastic application for the **Postdoctoral Position in Artificial Intelligence** within the **College of Computing (COLCOM)** at **Mohammed VI Polytechnic University (UM6P)**. As a final-year PhD candidate in Computer Vision and Machine Learning at the University of Ghana (dissertation defence scheduled for October 2026), my doctoral research has centered on **inference-time foundation-model composition**, **open-vocabulary spatial perception**, **multimodal reasoning**, and **probabilistic calibration under extreme data scarcity** ($\Delta\theta = 0$).

UM6P's College of Computing represents one of the most ambitious and transformative research environments in the Global South, bridging fundamental computing breakthroughs with high-impact applications in **agriculture, remote sensing, and multimodal knowledge systems**. I am eager to bring my expertise in multi-view geometric consensus, vision-language model alignment, and supercomputing-scale pipelines to COLCOM, leveraging the premier **TOUBKAL** supercomputer to advance resilient, label-efficient AI systems across Africa.

Research Alignment with COLCOM: Remote Sensing, Agriculture & Multimodal AI

My doctoral research directly addresses the core computational bottlenecks targeted by COLCOM's artificial intelligence and remote sensing initiatives:

1. **Open-Vocabulary Multi-View Perception for Drone and Satellite Remote Sensing:** Monitoring complex agricultural landscapes and environmental changes via multi-spectral drone and satellite feeds requires interpreting highly varied, uncurated imagery without demanding hundreds of thousands of manually annotated masks. In my research (accepted at **ACCV 2026** and under review at **WACV 2027** and **ICLR 2027**), I developed training-free open-vocabulary segmentation architectures that reconcile dense visual embeddings with 2D promptable segmenters (SAM 2.1) using projective camera geometry (K, T), ray casting, and non-parametric spatial consensus. On dense spatial benchmarks ($\sim$50M points across 312 scenes), my methods achieved state-of-the-art harmonic mean accuracy without altering foundation model weights ($\Delta\theta = 0$). This framework maps directly to agricultural remote sensing: fusing aerial drone perspectives with satellite imagery to achieve robust, zero-shot parcel segmentation, crop canopy classification, and multi-temporal land-cover tracking.

2. **Self-Supervised Multimodal Reasoning & Label-Free Process Rewards:** A major focus at COLCOM is developing multimodal AI, Large Language Models (LLMs), and Vision-Language Models (VLMs) tailored for agronomic knowledge and decision support. In my recent work (**SACAIR 2026 Submission**), I formulated physical consistency across independent viewpoints as an intrinsic, label-free process reward for multimodal reasoning models (**Qwen2.5-VL-7B**). By implementing an online Group Relative Policy Optimization (**GRPO/VERL**) pipeline on **vLLM**, I demonstrated that visual cross-view agreement forces reasoning models to ground their deductions in verifiable spatial facts, eliminating spatial hallucination and area-bias specification gaming ($\rho > 0.72 \rightarrow 0.334$) without requiring expensive human reward labels. At UM6P, I plan to extend this approach to agricultural VLM agents, enabling reliable agronomic question answering,

symptom diagnosis from crop imagery, and automated synthesis of open-source agricultural knowledge bases.

3. **Reliability Calibration & Anomaly Detection under Severe Domain Shifts:** Operational agricultural deployment demands trustworthy uncertainty quantification: models must reliably recognize unseen plant diseases, water stress, or unfamiliar soil conditions rather than outputting overconfident false positives. In my targeted **IEEE TPAMI** theoretical work (*RCASA*), I proved class-wise bounded updates and probability simplex preservation by uniting Platt confidence calibration with Bernoulli uncertainty estimation. This provides rigorous statistical guarantees for safe, calibrated autonomous monitoring in dynamic outdoor agricultural environments.

HPC Scaling on TOUBKAL & High-Impact Scientific Output

Conducting high-level research at COLCOM requires translating rigorous mathematical formulations into high-throughput computing implementations. Across my doctoral research, I have engineered memory-bounded evaluation pipelines utilizing float16/bfloat16 activation caching, multi-GPU parallelism, and chunked processing to evaluate massive datasets under strict hardware constraints.

The presence of the **African Supercomputing Center (ASCC)** and the **TOUBKAL supercomputer** at UM6P provides an unparalleled platform. I am prepared from day one to scale distributed foundation model inference, large-scale VLM reinforcement learning pipelines, and massive spatial coordinate lifting across multi-node GPU clusters. My objective at COLCOM is to publish prolifically in flagship computer vision, machine learning, and remote sensing venues—including **CVPR, ICCV, ECCV, ICLR, NeurIPS, IEEE TPAMI, and IEEE TGRS**—while releasing open-source benchmarks and foundation models that establish COLCOM as the global benchmark leader for African remote sensing.

Mentorship, Proposal Writing & Pan-African Academic Leadership

Beyond independent research, I am deeply committed to the academic mission of UM6P:

- **Mentorship & Student Supervision:** Having lectured in machine learning and data structures at Accra Technical University and mentored undergraduate researchers at the University of Ghana, I look forward to actively co-supervising PhD and master's students at COLCOM, guiding them through paper writing, experimental rigor, and submission to top-tier conferences.
- **Grant Writing & Research Proposals:** Having founded and led EED Soft Consult, I possess proven experience translating technical concepts into actionable project roadmaps. I am eager to actively contribute to competitive international grant proposals (e.g., Horizon Europe, African Union, and international agricultural research consortia) to secure sustained funding for COLCOM research initiatives.
- **Pan-African Scientific Bridge:** As a researcher trained at Ghana's premier university, joining UM6P represents a powerful opportunity to strengthen South-South research collaboration, fostering a unified Pan-African nexus of machine learning and computing excellence.

My complete application package—including my tailored CV, Research Statement, List of Publications, Doctoral Dissertation synopsis, and certified transcripts—is attached. I welcome the opportunity to discuss with the selection committee how my technical expertise, systems rigor, and research vision will contribute to the ongoing success of the College of Computing at UM6P.

Thank you for your time, consideration, and leadership in advancing African computing.

Sincerely,

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